Geavanceerde computerarchitectuur
Lab sessions

(JLI1QS)

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1 Session 1 - Getting Acquainted with GPU Programming

In this lab session, our primary objective was to familiarize ourselves with GPU programming. Unlike CPU programming, which prioritizes sequential thread execution, GPU programming excels in executing multiple threads in parallel. Our focus was on understanding the structure of a program and how it can be optimized for GPU execution.
We began by examining how a program can be broken down into smaller sequential tasks, or threads. To expedite execution, these threads can be run in parallel on the GPU. Threads are organized into blocks, with each block being processed by a distinct GPU core. In this session, our task was to reverse the order of elements in an array.
Initially, we attempted this using only one block containing nine threads. With this setup, all threads had to execute sequentially, leading to nine execution cycles. Next, we divided the threads into three blocks, each containing three threads. This allowed three threads to execute simultaneously, reducing the total execution cycles to three. This can be seen in figure 1.
Finally, we further divided the threads into nine blocks, with one thread per block. This approach necessitated only a single execution cycle, resulting in significant time savings. Although the impact was minimal in this particular case, the advantage would become more pronounced with a substantial increase in the number of threads to be executed.

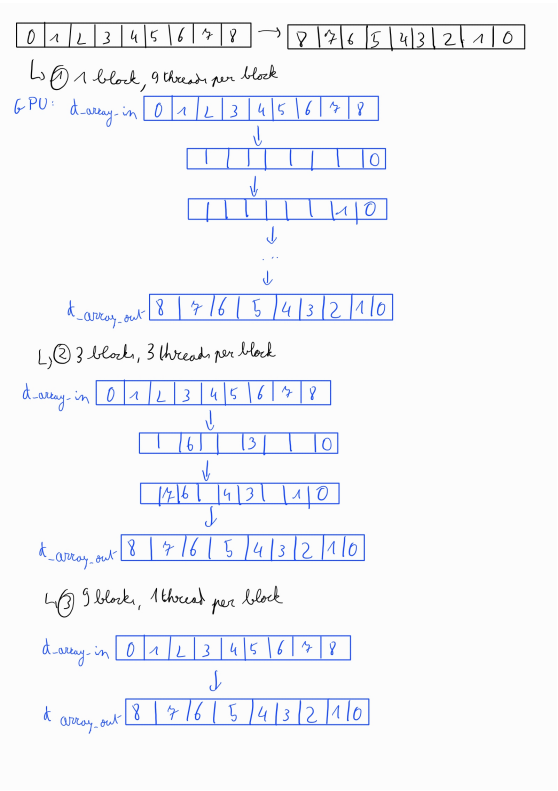


Figure 1: Illustration of blocks and threads

2 Session 2 - Race conditions